CLADE: Category-Level Answering from Deficient Evidence in Retrieval-Augmented Language Models

Anonymous submission

Abstract

Retrieval-augmented systems often retrieve passages that are accurate and relevant but state the answer at a coarser level of granularity than the question requires. For example, a question may ask which specific disease a drug treats (*familial hyperlipidemia*), while the retrieved passage mentions the broader disease family (*hyperlipidemia*). Prior work on context sufficiency treats such coarse passages simply as insufficient. We argue that they form a distinct case, which we call *resolvable insufficiency*: text alone cannot settle the fine-grained answer, but the broader category it mentions maps to a short list of candidate answers in a domain ontology. When language models read these coarse, *family-only* passages, they exhibit a phenomenon we call CLADE (Category-Level Answering from Deficient Evidence): they answer instead of abstaining or searching further. On biomedical questions from PrimeKG, family-only passages reduce abstention by 13.8 points compared to uninformative passages, and the resulting guesses are wrong about half the time. A counterfactual test confirms the cause: replacing an exact disease name with its broader family name lowers abstention by 17.4 points compared to deleting the sentence. Simple fixes like confidence filtering, scaling to a larger model, and explicitly instructing the model to abstain cannot resolve CLADE, and this replicates on ecological species interactions in GloBI. Finally, we test whether an ontology can turn coarse passages into exact answers. Given the correct family, an ontology list includes the true answer in 98.5% of cases, but string matching finds that family on the page for only 19.8%. Letting the model choose the family raises this to 30.0% and end-to-end recovery from 10.0% to 26.8%. Overall, our results show when language models should stop guessing and consult an ontology, and highlight which steps of that process remain unsolved. Codes are available at: https://anonymous.4open.science/r/CLADE.

1 Introduction

Retrieval-augmented generation (RAG) systems answer questions by reading retrieved text (Lewis et al. 2020). Recent work asks what a model should do when that text does not contain the complete answer. The usual framing is binary: a passage is either sufficient or insufficient (Joren et al. 2025), and a reliable model answers in the first case and abstains in the second (Wen et al. 2025). In practice, many retrieved passages fall between these two extremes. Consider the question *What disease does lovastatin treat?*, where the true answer is the specific condition *familial hyperlipidemia*. Suppose the retriever returns a passage stating that *lovastatin is used to treat hyperlipidemia* (Figure 1). This passage is accurate and relevant. It rules out every disease outside the hyperlipidemia family, but it is too coarse to specify which exact subtype is meant. In this case, a system can guess a specific disease, abstain, or look further using a medical ontology that lists specific members of the hyperlipidemia family.

We call this phenomenon *resolvable insufficiency*. The passage is insufficient on its own because it states the answer at a coarse category level rather than the required fine-grained entity level. Yet it is resolvable because the coarse name maps directly to a short, finite set of candidate answers in a structured ontology. When a model instead guesses a fine-grained answer directly from such coarse text, we name the behavior CLADE (Category-Level Answering from Deficient Evidence). Although this setting occurs frequently in real text—biomedical documents state the exact disease name for only a small fraction of knowledge-graph facts while describing many others at the family level—prior work overlooks how coarse evidence affects model abstention. Rather than proposing a surface-level patch, this paper digs into the mechanics of this phenomenon to understand how coarse evidence misleads large language models (LLMs) and where a neural reader should hand off to symbolic knowledge.

To study this systematically, we group retrieved passages into three categories based on how specifically they describe the target answer: *named* (exact-match) passages, which explicitly mention the fine-grained target entity or an exact synonym; *family-only* (coarse) passages, which omit the specific answer but mention a broader parent category (such as a disease family or biological genus) within two levels above the answer in an ontology; and *silent* (unmentioned) passages, which discuss the query subject (e.g., the drug) but mention neither the specific answer nor its broader family. Using this framework, we investigate three research questions: (1) how often reference texts state fine-grained biomedical facts versus coarse categories (and whether a coarse name helps a model on its own), (2) whether family-only passages cause models to stop abstaining and guess inaccurately, and (3) whether a domain ontology can turn a coarse family mention

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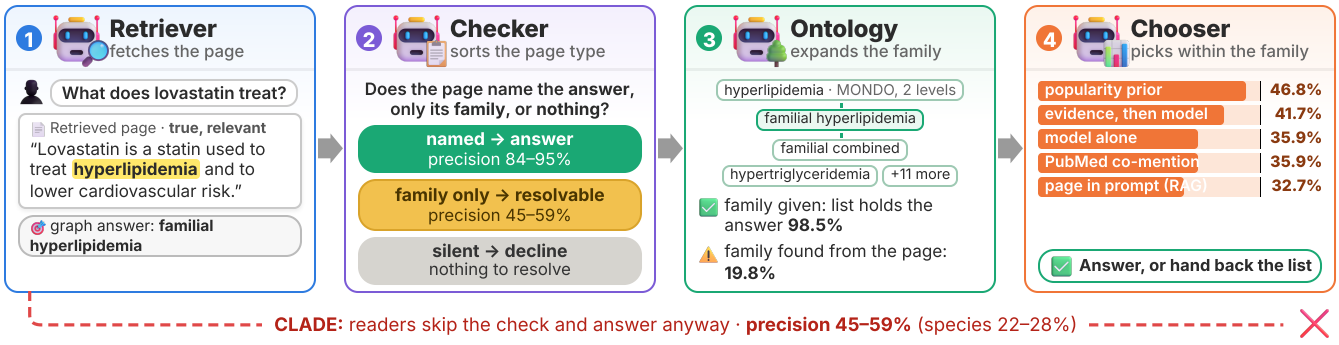


Figure 1: *Resolvable insufficiency* as an agent workflow, with the bottleneck we measure at each stage. The retrieved passage is accurate and relevant but names only the answer's family. A page-type check should send it to an ontology, which expands the family into a short candidate list for a chooser. The dashed red path is CLADE: the model guesses from the family-only passage instead. Today only the ontology step works reliably.

into the exact fine-grained answer. Our main contributions are as follows:

- We define *resolvable insufficiency* and CLADE, and measure how text states 1,800 PrimeKG facts and 23,694 GloBI interactions. Text omits the exact disease for 77.5–84.9% of PrimeKG facts, and a family name alone gives the model no closed-book advantage.
- On 856 PrimeKG questions and three 7–8B models, family-only passages cut abstention by 13.8 points relative to silent passages. About half of the extra answers are wrong, and precision falls from 84–95% to 45–59%.
- Counterfactual edits show that the family name causes this drop (−17.4 points against deleting the sentence). Confidence filtering, a 14B model and an instruction to abstain do not stop CLADE, and the effect replicates on 900 GloBI questions. In free text, no model gives the exact answer from a family-only passage.
- On 410 family-only facts, an ontology list holds the answer 98.5% of the time once the family is known, but string matching finds the family from the passage only 19.8% of the time. A model that chooses the family raises this to 30.0% and end-to-end recovery from 10.0% to 26.8%, and a popularity prior beats the model at choosing within the list.

2 Related Work

Insufficient context. Retrieval failures are a common weakness in RAG systems (Barnett et al. 2024). Joren et al. (2025) study the errors through the lens of context sufficiency. They label retrieved passages as either sufficient or insufficient and find that LLMs often hallucinate instead of abstaining when context is insufficient. Although they note that a passage can narrow down the answer without fully settling it, they group all such cases under insufficient context. Prior work also shows that models are easily thrown off by irrelevant text (Shi et al. 2023; Yoran et al. 2024), and Setiawan (2026) show that misleading passages cause prompt-based abstention to fail, whereas missing passages do not. Our focus, *resolvable insufficiency*, differs from both irrelevant and misleading context. A coarse passage is factually accurate and points in the right direction, yet this partial relevance is enough to trigger CLADE and break abstention.

Partial evidence. To handle imperfect passages, several methods train or prompt models to grade evidence quality. Self-RAG (Asai et al. 2024) trains a model to judge whether a passage fully or partially supports a generated answer, and ALCE (Gao et al. 2023) checks whether citations back up model claims. Similarly, CRAG (Yan et al. 2024) scores retrieved documents to trigger new searches when support is weak, and SURE-RAG (Qiu, Han, and Huang 2026) checks whether context is sufficient before answering. While these methods detect that support is partial, they do not identify what kind of information is missing. A coarse mention is a structured form of partial evidence: it shows that fine-grained entity detail is missing and indicates where to find the candidate answers in an ontology.

Abstention. When evidence falls short and cannot be resolved, a reliable model should abstain. Recent surveys and benchmarks show that models often guess when they should hold back (Wen et al. 2025; Kirichenko et al. 2025), including in clinical settings (Machcha et al. 2026). Most existing methods focus on detecting when a model lacks knowledge so it can decline to answer (Kadavath et al. 2022; Feng et al. 2024). While those studies measure whether a model abstains overall, we test how the granularity of retrieved evidence (*named*, *family-only*, *silent*) causes a model to stop abstaining and guess.

Ontologies in RAG. Rather than guessing or abstaining on a coarse passage, a system can consult a structured ontology. Recent RAG systems use symbolic structures to guide retrieval: OG-RAG (Sharma, Kumar, and Li 2024) grounds retrieval in domain ontologies, and TaSR-RAG (Sun et al. 2026) uses taxonomies for multi-hop retrieval. Broader work on combining LLMs with knowledge graphs and symbolic reasoning is surveyed by Pan et al. (2024) and d'Avila Garcez and Lamb (2023). Unlike pipelines that query an ontology on every question, we study the specific case where a coarse passage calls for an ontology lookup. We also measure each

	PrimeKG	GloBI
Source records	8,100,498	7,822,670
Edges	indication 9,388 contraind. 30,675 off-label 2,568	eats 235,960 preys on 87,678 parasite of 117,894 pollinates 15,854
Hierarchy	MONDO	GloBI taxonomy
child–parent links	17,801	77,852
families, ≥2 members	3,182	8,370
median / max size	3 / 150	3 / 315
Text	Wiki. + PubMed	Wikipedia
sources with text	574 drugs	5,425 species
facts measured	3,444	23,694
Questions	856	900

Table 1: Data for both domains. PrimeKG links drugs to diseases, and MONDO groups diseases into families. GloBI links species to species, and its taxonomy groups species into genera. A *family* is a parent with two or more children. *Facts measured* are edges we check for a mention of the answer.

step of the pipeline: extracting the family, listing its members, and picking the final answer; then we check where the process breaks down.

Fact coverage in text. Our analysis also connects to work on what reference and pretraining texts actually state. Mallen et al. (2023) and Kandpal et al. (2023) link a model's closed-book knowledge to entity popularity and document counts, while LMEnt (Gottesman et al. 2026) tracks entity mentions across pretraining data. Whereas those studies count mentions of individual entities, we measure how often reference texts state relational facts at an exact (*named*) level versus a coarse (*family-only*) level.

Fine-grained biology. Biology shows the same gap between coarse and fine answers. On Bio-VQA, taxonomic accuracy falls from 0.72 at phylum to 0.12 at species (Rayeed et al. 2027). And models lose accuracy from genus to species (Rayeed et al. 2025) even on curated specimen images with validated traits (Rayeed et al. 2026). Knowledge-based VQA benchmarks also contain answers that their corpus does not support (Ma et al. 2026). We investigate the text side of this gap: a page that names the genus or family but not the answer.

3 Data and Problem Setup

Knowledge graphs and ontologies. We use drug-disease relations from PrimeKG, linked to the MONDO Disease Ontology. We reconstruct MONDO into 17,801 edges across 11,788 diseases (excluding parents with >400 descendants). To test generalizability beyond medicine, we evaluate species interactions from GloBI, using 457,386 species-to-species edges across four relations (*eats*, *preys on*, *parasite of*, *pollinates*) with GloBI's taxonomy (species, genus, family) as the hierarchy. Table 1 summarizes both domains.

Reference text corpora. For PrimeKG drugs, we use English Wikipedia articles and PubMed abstracts. For GloBI organisms, we use species' Wikipedia articles (dropping stubs <3000 characters). All articles are stripped of markup, templates, infoboxes, tables, and file links.

		Text names it		Text does not	
Relation	*n*	knows	not	knows	not
Indication	600	9.7	7.2	21.8	61.3
Contraind.	600	1.5	5.2	20.7	72.7
Off-label	600	4.3	6.2	24.2	65.3
All	1,800	5.2	6.2	22.2	**66.4**

Table 2: Share of PrimeKG facts (%) by whether the drug's Wikipedia article names the disease and whether Qwen2.5 0.5B picks it from five options closed book ("knows").

Relation	Text	*n*	Named	Family	Silent
Indication	Wikipedia	1,722	20.3	15.5	64.2
	PubMed	1,722	26.3	18.6	55.1
	both	1,722	29.0	19.7	51.3
Contraind.	Wikipedia	1,722	9.8	4.1	86.1
	PubMed	1,722	11.4	5.4	83.2
	both	1,722	15.9	6.9	77.2
PrimeKG	both	3,444	22.5	13.3	64.2
Eats	Wikipedia	8,000	4.0	6.6	89.3
Preys on	Wikipedia	8,000	2.2	4.4	93.4
Parasite of	Wikipedia	2,389	12.5	15.5	72.0
Pollinates	Wikipedia	5,305	2.2	8.0	89.7
GloBI	Wikipedia	23,694	3.9	7.1	89.1

Table 3: Share of knowledge-graph facts whose text names the answer, names only its family, or names neither. PrimeKG uses 574 drugs with three facts of each relation.

Evidence granularity. We classify retrieved text into three levels: *named* (exact name, synonym, abbreviation), *family-only* (broad parent category within two levels), and *silent* (neither). A control with unrelated disease pairs shows that matches track facts, not the size of the name set (unrelated matches occur in only 0.9–1.4% of Wikipedia articles, versus 6.7–19.3% for true diseases).

Corpus coverage. Evaluating 1,800 PrimeKG facts with Qwen2.5 0.5B (closed book) reveals that for 66.4% of facts, the text lacks the exact disease name and the model lacks the fact. As detailed in Table 2 and Table 3, the exact disease is missing in 84.9% of Wikipedia articles, 81.1% of PubMed sets, and 77.5% combined, with random guessing accounting for most closed-book correct answers on unnamed facts.

Closed-book recall. Although family-only text covers 13.3% of PrimeKG facts, testing on Qwen2.5-7B shows that coarse family mentions do not reflect stronger underlying model knowledge: it answers 39.6% of family-only facts, 32.0% of silent facts and 65.7% of named facts correctly (Appendix B). Odds ratio (OR) compares the odds of a correct closed-book answer in two groups. OR=1 means no difference, and OR<1 means the first group is answered correctly

less often. We pool the relations with the Mantel–Haenszel estimator; p is the chance of a gap this large if the groups did not differ. Family-only facts are answered far less often than named facts (OR = 0.27, $p = 9 \times 10^{-9}$) and as often as silent facts (OR = 1.10, $p = 0.61$). While an exact Wikipedia mention yields a 3.7× increase in the odds of model knowledge, a family name provides no measurable advantage on its own. This motivates testing how models handle coarse partial evidence at inference time, and whether symbolic ontologies can recover exact answers.

4 Methodology

We run four experiments. We measure how family-only pages change abstention, edit pages to isolate the effect of the family name, test three common fixes, and test whether an ontology can recover the exact answer.

4.1 Abstention Evaluation Setup

Questions. We curate 856 multiple-choice questions from the PrimeKG indications and contraindications: 300 with a *named* page, 256 with a *family-only* page, and 300 with a *silent* page. Each relation and group has 150 questions, except family-only contraindications, where only 106 facts exist. Each question has five candidate diseases and a sixth option, *Not enough information to tell*. The four wrong options come from the answer's MONDO neighborhood: siblings first, then cousins, then random diseases only if both run out. We exclude the answer's ancestors and descendants and any other disease the drug has under the same relation.

Pages. We ask each question twice: closed book and with a page. The page starts with the drug's cleaned Wikipedia article, followed by its PubMed abstracts. We keep the first 800 characters and a 280 character window around each mention of the answer or its family, merge overlapping windows, and cap the page at 1800 characters. A silent page has no mentions, so it is the article's opening.

Models and Scoring. We evaluate five instruction-tuned open-weight models from three families: Qwen2.5 0.5B, 7B and 14B (Qwen Team 2024), OLMo 2 7B (Team OLMo et al. 2024), and Granite 3.1 8B (IBM Granite Team 2024). The model sees the page, the question and options A–F. We take the letter with the highest next-token probability, renormalized over the six letters. Choosing *Not enough information to tell* counts as a decline. As a format check, we also ask 120 matched questions with no options and score each generated answer by name matching.

Controls. Two differences between groups could bias the comparison. First, family-only answers more often have enough relatives to fill all four wrong options, which makes those questions harder. Our main analysis therefore keeps only questions whose four wrong options all come from the neighborhood: 106 named, 215 family-only and 160 silent. Appendix D reports all 856 questions, with the same pattern.

Second, the groups hold different facts, which may differ in difficulty before any page is read. We therefore compare each question with itself. For a question q and an outcome such as a decline, let $y_q^{\text{page}} = 1$ if the outcome occurs when the model reads the page and 0 otherwise, and let y_q^{closed} be the same without the page. The *page effect* of q is $y_q^{\text{page}} - y_q^{\text{closed}}$. Our main measure is the difference in mean page effect between the family-only questions F and the silent questions S:

$$\Delta = \frac{1}{|F|} \sum_{q \in F} \left(y_q^{\text{page}} - y_q^{\text{closed}}\right) - \frac{1}{|S|} \sum_{q \in S} \left(y_q^{\text{page}} - y_q^{\text{closed}}\right). \quad (1)$$

For declines, $\Delta < 0$ means a family-only page pushes the model toward declining less than a silent page does. To pool the 7–8B models, we average each question's page effect over the models first, so each question counts once.

Statistics. Effects are in percentage points. Brackets give 95% confidence intervals (CI) from 5,000 bootstrap resamples of questions; an interval that excludes 0 marks a reliable effect. p-values come from two-sided z-tests (Appendix C).

4.2 Counterfactual Evaluation

Page edits. The groups in Section 4.1 test different facts. To hold the fact fixed, we take 130 named questions whose answer has a MONDO parent. Then we edit each page in two ways: *family edit* replaces every mention of the answer with its parent's name, and *removed edit* deletes every sentence that mentions the answer (Figure 3). The question, options and model stay the same.

Simple fixes fail. We test three ways to curb CLADE. *Confidence filtering* uses the renormalized probability of the chosen option; we report the AUROC for separating correct from wrong answers. *Scaling* compares Qwen2.5 at 0.5B, 7B and 14B. *Instruction* adds a line to the prompt telling the model to choose *Not enough information to tell* when the page names only a broad category.

Second domain. We repeat the main experiment and counterfactual edits on 900 GloBI questions: 75 per page type in each of four relations. Wrong options come from the same genus first, then the same family.

4.3 Ontology Recovery

We test recovery on 410 PrimeKG facts whose Wikipedia and PubMed text names only the disease family (Figure 1). A drug often has several true diseases in a family (2.47 per list on average), so a pick counts as correct if it matches any of them. Recovery has three stages:

1. **Build the list.** The oracle *given-family* list takes the narrowest MONDO ancestor named in the text and lists its descendants up to two levels down, capped at 64. The automatic *page-built* list takes every MONDO parent named in the drug's Wikipedia article and joins their descendants; lists over 64 count as too broad. The *model-chosen* list asks Qwen2.5-7B which of the families on the page (up to ten) holds the answer and expands only that one.
2. **Choose one.** We compare five options: (1) Qwen2.5-7B closed book, (2) Qwen2.5-7B with the page in the prompt, (3) a reverse lookup to check whether the candidate's own Wikipedia article names the drug, (4) PubMed co-mention counts with relation keywords, raw and normalized by the disease's PubMed volume, and (5) a *popu-*

Model	Passage Type	Correct	Wrong	Declines	Prec.
Qwen2.5 7B	named	71.7	3.8	24.5	95.0
	family-only	26.5	18.1	55.3	59.4
	silent	13.8	2.5	83.8	84.6
OLMo 2 7B	named	77.4	15.1	7.5	83.7
	family-only	37.2	44.7	18.1	45.5
	silent	30.6	24.4	45.0	55.7
Granite 3.1 8B	named	72.6	7.5	19.8	90.6
	family-only	30.2	26.5	43.3	53.3
	silent	15.0	10.0	75.0	60.0
Qwen2.5 0.5B	named	77.4	14.2	8.5	84.5
	family-only	34.4	55.8	9.8	38.1
	silent	28.7	50.6	20.6	36.2
Qwen2.5 14B	named	67.9	2.8	29.2	96.0
	family-only	24.2	17.2	58.6	58.4
	silent	11.9	2.5	85.6	82.6

Table 4: Open-book outcomes (%) on PrimeKG questions where all four distractors come from the answer's MONDO neighborhood (106 named, 215 family-only, and 160 silent questions). Precision is the percentage of correct answers among questions where the model did not abstain.

larity prior that picks the disease with the most PubMed abstracts, ignoring the drug.

3. **Compare without ontology.** Two baselines replace the family list with a list of the same length that holds the answer plus random diseases, scored by the model with or without the page. We test paired differences with McNemar's test (McNemar 1947).

5 Results

5.1 Coarse evidence weakens abstention (CLADE)

Models abstain less with partial evidence. Table 4 shows outcomes when the model reads the page. All 7-8B models abstain much less on a family-only page than on a silent page: Qwen2.5-7B from 83.8% to 55.3%, OLMo 2 from 45.0% to 18.1%, and Granite from 75.0% to 43.3%. Many of the extra answers are wrong. Precision, the share of answers that are correct, falls from 83.7-95.0% on named pages to 45.5-59.4% on family-only pages.

CLADE drop comes from context. Family-only facts are harder even closed book: they draw more wrong guesses than silent facts in all three 7-8B models (Appendix D). The page effect in Eq. 1 removes this difference. Pooled over the models, a silent page raises abstention by 28.5 points over closed book, but a family-only page raises them by only 14.7 points. The difference Δ is -13.8 points $[-20.3, -7.5]$, and it stays significant after a Bonferroni correction for three outcomes (p=1.1×10^{-4}; Figure 2). The freed answers split between wrong answers (+7.7 [1.2, 14.0]) and correct ones (+6.1 [−0.1, 12.4]). In plain terms, a silent page makes the models stop and say they cannot tell; a family-only page makes them stop much less often, and they guess instead. The drop is significant for OLMo 2 (-22.1, p=6×10^{-6}), Granite (-18.2, p=6×10^{-4}) and Qwen2.5-0.5B (-8.4, p=0.027). Qwen2.5-7B shows almost no gap (-1.1, p=0.82). It already declines family-only questions far less often closed book (32.1% vs. 59.4%), and the page raises its abstention by about the same amount in both groups.

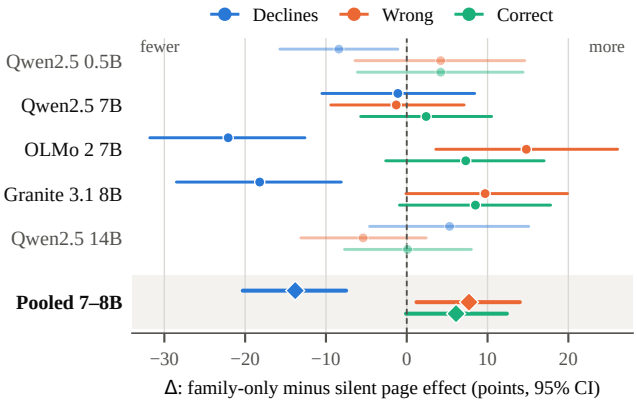


Figure 2: Effect of a family-only page relative to a silent page. Each point is Δ from Eq. 1 for one outcome, with its 95% bootstrap CI; points left of zero mean the family-only page lowers that outcome. The pooled row averages the three 7-8B models over 481 questions. Faded rows are models outside the pooled set.

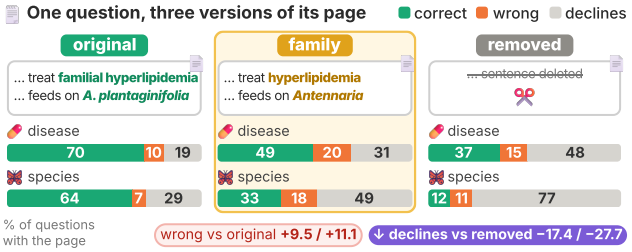


Figure 3: Page edits on the same questions: PrimeKG (130) and GloBI (255). The *family* edit replaces the answer with its parent; the *removed* edit deletes the sentence. Bars show outcomes with the page (%), averaged over the 7-8B models.

An example. For moxifloxacin, PrimeKG lists *Klebsiella pneumonia*, and the page says only that the drug treats pneumonia. All five models pick some bacterial pneumonia with confidence up to 1.00, and none abstains. For lovastatin, OLMo 2 is right closed book but picks a sibling after reading a page that says only *hyperlipidemia* (Appendix I).

Family size. We split the family-only questions into thirds by family size. The drop is large for families of up to 10 members (-17.3 [-26.2, -8.2]) and of 11-20 members (-18.4 [-26.6, -10.3]). It shrinks to -6.5 [-15.7, $+3.0$] for families above 20 members (Figure 4c). A narrow family makes the page look more answerable.

5.2 Counterfactual Edits confirm CLADE

The edits change only the page (Figure 3). Against the original page, the family edit lowers correct answers by 21.0 points and raises wrong answers by 9.5 points, while abstention rises by only 11.5 points. The key comparison is with the removed edit. Keeping the family name instead of deleting the sentence cuts abstention by 17.4 points [−23.6, −11.5] and raises wrong answers by 5.1 points [0.5, 10.3]. Every model shows this order: from removed to family, abstention falls from 70.0% to 43.8% for Qwen2.5-7B, from 56.2% to 40.8% for Granite, and from 18.5% to 7.7% for OLMo 2. The family name alone suppresses abstention.

5.3 Three Fixes Fall Short

Confidence does not flag wrong answers. AUROC measures how well confidence ranks the correct answers above wrong ones: 1.0 is perfect and 0.5 is chance. On family-only pages it is 0.59-0.72, against 0.73-0.87 on named pages, in every model (Figure 4a). Wrong guesses are confident: their mean probability is 0.77-0.99 in 7-14B models. A threshold that keeps correct answers also keeps most wrong ones.

Scaling does not help. Qwen2.5-14B has the best precision on named pages (96.0%). On family-only pages, precision rises from 38.1% at 0.5B to 59.4% at 7B, but then stops: 14B reaches 58.4%, and it gives 14.7 points more wrong answers than on silent pages (p=6×10^{-6}).

Instruction. Telling the model not to answer when the page names only a broad category raises abstention by 15.2 points on family-only pages (Figure 4b). It also raises them by 14.0 points on silent pages and 9.1 points on named pages, where it costs 6.3 points of correct answers. Precision on family-only pages stays at 47.9-64.9%, far below named pages (89.6-94.7%). The models scores change differently, OLMo 2 abstains far more on all page types, while Granite barely changes (Appendix F). However, none of these models tells a family name from a specific answer.

5.4 Second Domain: GloBI

Every finding on PrimeKG replicates on species interactions (Table 5). A silent page raises abstention by 12.1 points, but a page that names only the genus or family lowers them by 2.3 points (Δ=−14.3, p=1.4×10^{-9}). The effect is largest in OLMo 2 (−31.7). Here the extra answers are mostly wrong (+11.6 wrong, p=2.5×10^{-8}; +2.7 correct, p=0.09). Precision on family-only pages falls to 22-28%, close to the 20% chance rate, because sibling species are harder to tell apart than sibling diseases. The page edit is even sharper: on 255 questions, keeping the genus name instead of deleting the sentence cuts abstention by 27.7 points [−32.2, −23.5] and raises wrong answers by 6.7 points (Figure 3). CLADE comes from hierarchical evidence in text, not from medical terms (Appendix G).

5.5 Ontology Recovery

List coverage. When the right family is given, its MONDO descendants form a list of median 14 candidates. The list holds the sampled answer 98.5% of the time and some true answer 99.5% of the time (Table 6).

Metric	PrimeKG	GloBI
Questions in pooled set (all-neighbor)	481	833
Abstention page effect, family − silent	−13.8	−14.3
Wrong-answer page effect, family − silent	+7.7	+11.6
Precision on named passages (%)	84–95	84–95
Precision on family-only passages (%)	45–59	22–28
Confidence AUROC on named passages	.73–.87	.82–.86
Confidence AUROC on family-only passages	.66–.72	.52–.68
Edit: abstention, family − removed	−17.4	−27.7
Edit: wrong answers, family − removed	+5.1	+6.7
Edit: correct answers, family − original	−21.0	−30.8

Table 5: Comparison of biomedical drug–disease facts (PrimeKG with MONDO) and ecological species interactions (GloBI with biological taxonomy) across the three 7–8B models. Effects are reported in percentage points; ranges span individual models. Every difference reported has a 95% CI excluding zero.

Metric	Given	Page-built	Model-chosen
Usable lists (of 410)	410	166	375
Holds sampled answer	98.5%	19.8%	30.0%
Holds any true answer	99.5%	24.6%	44.9%
Median list size	14	32	12
End-to-end recovery	–	10.0%	26.8%

Table 6: Candidate lists for the 410 family-only facts: given family (oracle), all families on the page (page-built), and the one family Qwen2.5-7B picks from the page (model-chosen). Of the 244 unusable page-built lists, 209 exceed the 64-candidate cap and 35 match no multi-child MONDO family. End-to-end recovery uses the best chooser for each list.

Finding the family. When the family must come from the drug's Wikipedia page, usable lists hold the answer for only 19.8% of facts. The family is rarely missing; it is buried. A drug article names a median of five MONDO categories, such as comorbidities, side effects and other uses, and their union exceeds 64 candidates for 209 of 410 facts. PubMed makes this worse: coverage drops to 3.7%. A model does better than string matching. When Qwen2.5-7B reads the page and picks one of the families it names, the chosen list holds the answer for 30.0% of facts (44.9% for any true answer), above the first-mentioned family (25.1%) and a random one (13.9%). The ceiling is 50.0%: for half of the facts, the right family is not among the usable families on the page.

Choosing a member. Within the given-family list (Figure 5), Qwen2.5-7B alone reaches 35.9% (32.4% with the page in the prompt), and raw PubMed co-mentions also reach 35.9%. Evidence first, with the model as a fallback, reaches 41.7%. Yet the best chooser ignores the drug: picking the disease with the most PubMed abstracts reaches 46.8%. When we divide co-mentions by each disease's abstract count, accuracy drops from 35.9% to 25.4%, so co-mentions work mainly because common diseases are studied more.

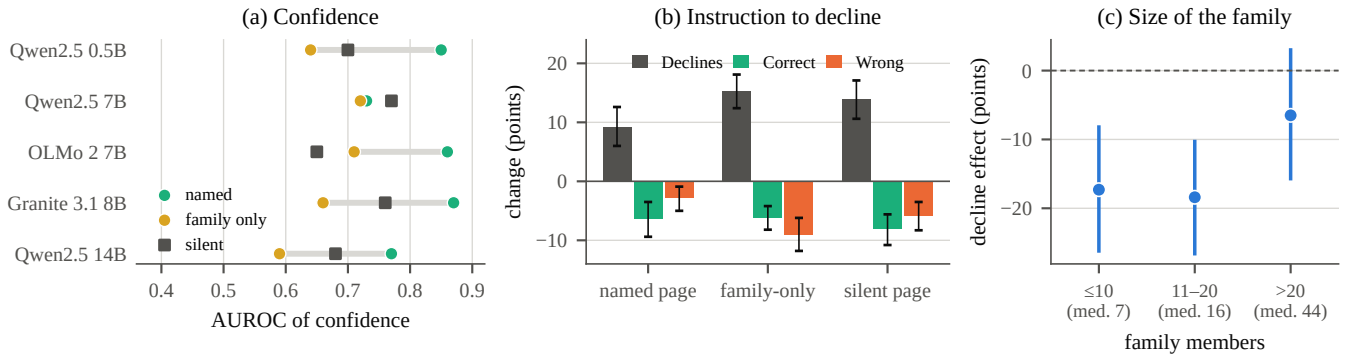


Figure 4: Three candidate fixes. (a) Confidence separates correct from wrong answers poorly on family-only pages. (b) An instruction to decline on broad categories raises abstention on every page type, including named pages (pooled 7-8B models, 95% CIs). (c) The drop in abstention is largest when the family has 20 or fewer members.

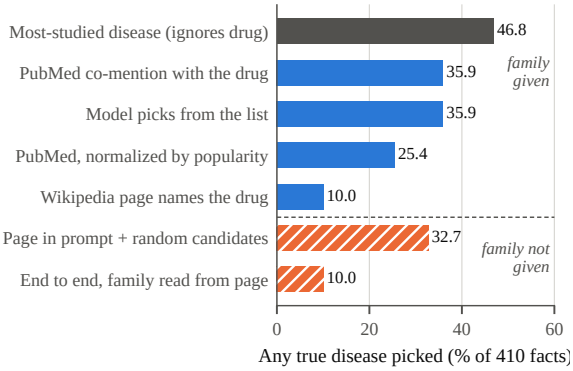


Figure 5: Accuracy of choosing the exact disease for 410 family-only facts. Blue and gray bars choose from the given-family list; hatched bars do not receive the true family.

Errors and end-to-end recovery. Most misses pick a better-studied sibling, such as *ovarian carcinoma* instead of *malignant germ cell tumor of ovary* for carboplatin, or a valid use missing from PrimeKG, so strict graph scoring understates accuracy (Appendix H). Without the given family, page-built lists recover the answer for 10.0% of facts with Wikipedia and 1.5% with PubMed added; model-chosen lists recover 26.8% (Table 6).

6 Discussion

Hand-off point. Our results mark a clear boundary between the language model and the ontology. On a family-only page, models take the family as license to guess. The ontology turns the same family into a short list that holds the answer, once the system knows which family on the page belongs to the question.

A three-step workflow. This suggests a simple workflow (Figure 1). (1) **Check the page type**, and answer from the page only if it names the answer. (2) **Find the target family** tied to the queried relation; if none fits, abstain. (3) **Expand and choose** with evidence and a popularity prior; answer only when one candidate clearly leads, otherwise return the list. Current models skip steps 1 and 2, and neither confidence nor instructions restore step 1. Step 2 is the main bottleneck: string matching finds the right family for 19.8% of facts, and a model chooser raises this to 30.0%, against a ceiling of 50.0%. Any chooser for step 3 should beat the popularity prior.

Beyond sufficiency. Recent work asks whether a page is sufficient (Joren et al. 2025); our results show that *how* it falls short matters. A family-only page differs from a silent one: it carries real signal, it breaks abstention, and it points into a symbolic hierarchy. Treating it as its own case lets a RAG system send partial evidence to an ontology instead of guessing or discarding it.

7 Limitations

We study two domains and run recovery only on PrimeKG. We use multiple-choice questions with an explicit decline option, which controls the wrong options and the scoring but may differ from free-text refusals. A free-text check points the same way (Appendix K): no model gives the exact answer from a family-only page, and 17.5–45.0% of its answers name the family itself. We test open models up to 14B, and the page edits cover only named facts whose answer has a parent. Page-built lists rely on exact string matching. Family-only contraindications are few (106 questions).

8 Conclusion

Retrieved text often states facts at a coarser level than the question asks. We show that this *resolvable insufficiency* triggers CLADE in medicine and ecology: a family or genus name leads models to stop declining and guess, which roughly halves precision. An ontology can narrow the answer to a short list that holds it 98.5% of the time once the right family is found, and a model that chooses the family from the page already raises end-to-end recovery from 10.0% to 26.8%. Finding the family more reliably and beating a popularity prior remain the key open problems for neuro-symbolic RAG.

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